

R65 Help Booklet

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animate

ANIMATE

```
func expaint:boolean;  
  Handles toggle graphics.  
  Provides code to paint and apply motion.  
  Returns true if animation loop should be  
  stopped., Stops if motion detects stop  
  (e.g. expaint:=false)
```

```
func exkey(ch:char):boolean;  
  Provides code to handle keys pressed.  
  Returns true if animation loop should be  
  stopped because of key pressed.  
  (e.g. exkey:=ch=ord(0);)
```

Usage:

```
{ $I IANIMATE:P }
```

```
animate(autorepeat)
```

Run the animation loop.
Set autorepeat to true if cursor key repetition
should start immediately, not after short
delay.

arglib

LIBRARY ARGLIB;

MEMORY

```
NUMARG    = $005f: integer&;
ARGLIST   = $0060: array[10] of integer;
ARGLISTS  = $0060: array[63] of char&;
ARGTYPE   = $00a0: array[31] of char&;
```

VARIABLES

```
_carg: integer;
```

ROUTINES

```
proc _argerror(e: integer);
proc _agetval(var value:integer; var default:boolean);
proc _agetstring(var string: array[15] of char;
  var default: boolean; var cyclus, drive: integer);
func option(opt:char):boolean;
```

bounce

BOUNCE – A Bouncing Ball Simulation

In this simulation, a ball bounces off the floor, ceiling, walls, and a circle. Gravity and bounce friction affect the motion of the ball.

The following modes are available:

1. No circle
2. Ball outside a small circle
3. Ball inside a large circle

The simulation starts with a random velocity.

The following keys control the simulation:

Up/Down	Increase/decrease vertical speed
Right/Left	Increase/decrease horizontal speed
Space	Select next mode
Escape	Quit simulation

cat

CAT filename [firstline]

display a text file, maximal 40 lines
firstline: skip lines until first line

checkbuild

CHECKBUILD – Check All Pascal Programs

This command checks whether all Pascal programs can be compiled without errors. It is especially useful after modifying a library.

Use as follows:

```
CD PSOURCE  
CHECKBUILD  
CD
```

A complete report is written to printout.txt.

The report is also displayed on the screen, but it will usually be larger than the terminal scroll buffer.

clean

CLEAN drive

drive is the disk drive to clean, default 1

- Deletes all except the latest version (cyclus) of each file
- Deletes remaining compiler temporaries

CLEAN does not free up any space. The deleted files remain on the disk. Use PACK to free up space after CLEAN.

compile

COMPILE name[.cy[,drv]] [/option]

name: file name of program to compile

option: p: no hard copy print

l: put line info in object code

r: range checking for index

n: no loader file

f: force compile

Default drive is 1.

The /f option forces a compilation even if there is a object file with equal or higher version on disk 0 or 1.

Compiler directives:

{ \$I filename } include file

{ \$R- } { \$R+ } range checking, default false

{ \$U- } { \$U+ } unique identifiers in scope

delete

```
DELETE name[.cy[,drv]] [/S]  
name: name of Pascal source file  
can include a subtype :X  
supports wildcards  
default drive is 1  
default for subtype is :P  
/S silent (no output on screen)
```

Delete one or several files

dir

DIR [drive] [name] [/s]

where	/S	Sort directory
	/F	Full directory with dates
	/D	Include deleted files
drive		Drive number, default 1
name		name of a disk

Examples	DIR	DIR /S
	DIR 0	DIR 0 /S
	DIR PSOURCE	DIR PSOURCE /S

edit

EDIT name[.cy[,drv]]

name: name of Pascal source file
can include a subtype :X

default drive is 1

default for subtype is :P

Edit a sequential file with the external editor.

errors

System Error Codes

=====

01	Read/write error
02	Checksum error
03	Escape exit during read/write
04	Record number error
05	File type error
06	File not found
07	Disk not ready
08	Directory full, file not stored
09	Illegal IRQ
10	Expression missing
11	Memory cell cannot be changed
12	Break table full, not inserted
13	Illegal memory cell for break
14	Double breakpoint setting
15	End of line expected
16	Syntax wrong in register name or =
17	Breakpoint not found in table
18	Syntax wrong in store
19	File subtype wrong or missing
20	Wrong file type, not runnable
21	Unknown monitor command
22	Illegal opcode for STEP/TRACE
23	Too many open files, not opened
24	Direction error in sequential read/write
25	Wrong file number, file not open
26	Disk full, file not stored
27	Random access index out of range
28	Illegal drive for random access
29	File not open for random access write

EXDOS Error Codes

=====

61	Wildcard not allowed
62	Only for disk, not tape
63	Illegal copy
64	File too large
65	Write error
66	Import error
67	Unknown emulator command
68	Unable to run external editor
69	File not changed or no new file

Pascal Runtime Error Codes

=====

81	Division by zero
82	Stack overflow
83	Index out of bounds
84	Program not found or not a Pascal program
85	Illegal P-code
86	Escape during execution
87	No loader file made by compiler
88	Heap overflow
89	Pointer not allocated (NIL)
90	Writing to constant string not allowed
91	String too long (more than 63 characters)
92	Only the last allocated string can be released

Pascal Argument Error Codes

=====

101	Argument is not string or default
102	Argument is not number or default
103	Argument is not starting with /
104	Unknown argument
105	Drive is not 0 or 1
106	Argument syntax error
107	Too many arguments

export

EXPORT name[.cy[,drv]]

name: name of Pascal source file
can include a subtype :X

default drive is 1

default for subtype is :P

Export a sequential file.

filelib

LIBRARY FILELIB;

MEMORY

```
FILFLG = $00da: integer&;
FILDRV = $00dc: integer&;
FILTYP = $0300: char&;
FILNAM = $0301: array[15] of char&;
FILCYC = $0311: integer&;
FILSTP = $0312: char&;
FILNM1 = $0320: array[15] of char&;
FILCY1 = $0330: integer&;
```

ROUTINES

```
func _uppercase(ch1: char): char;
proc _asetfile(name: array[15] of char;
    cycls,device: integer; subtype: char);
proc _runprog(name0: array[15] of char;
    cyc: integer; drv: integer);
proc _chainprog(name0: array[15] of char;
    cyc: integer; drv: integer);
proc _write_label;
func _bestmatch: boolean;
func _freedrv(drive:integer; printit:boolean);
```


graph

GRAPH filename

display the graph of function values
stored in table on internal display

TEKGRAPH filename

display the graph of function values
stored in table on Tektronix 4010

MAKETABLE filename

make a table for GRAPH/TEKGRAPH

graphics

LOW RESOLUTION GRAPHICS ON THE R65 COMPUTER

Graphics can be displayed in split mode in a graphic overlay, or in fullscreen mode.

To toggle between full screen and split mode use the key "ctrl-l".

To clear the graphics canvas and release the graphics memory use the command "GREND".

grep

GREP "pattern" filename

Find pattern in text files.

If quotes are used, pattern can contain lower case letters and other characters. Pattern without quotes can only use upper case letters and numbers.

Filename can include wildcards , cyclus and drive.

Default is drive 1.

help

HELP topic

where topic is a command name or another file with extention :H on drive 1 or 0.

ianimate

ROUTINES

```
proc animate(arepeat:boolean);
```

ichkesc

ROUTINES

```
func chkesc(checktoggle:boolean):boolean;
```

icircle

ROUTINES

```
proc circle(x,y,r,c:integer);
```

ifile

ROUTINES

```
proc runprog
  (name: array[15] of char;
   cyc: integer; drv: integer);
proc _writename(text: array[15] of char);
proc setsubtype(subtype:char);
func contains(t:array[7] of char):boolean;
func letter(ch:char):boolean;
proc setargs(name:array[15] of char;
  _carg0,cyc,drv:integer);
proc setargi(val,_carg0:integer);
```


import

```
IMPORT name[.cy[,drv]]  
  name:    name of Pascal source file  
           can include a subtype :X
```

```
default drive is 1  
default for subtype is :P
```

Import a sequential file

iotest

```
PROGRAM IOTEST
USES SYSLIB, STRLIB
VAR S1: CPNT
    FOUT: FILE
BEGIN
    WRITELN ('IOTEST')
    STRFIO('IOTEST:B',1)
    OPENW(FOUT)
    WRITELN(@FOUT, 'IOTEST')
    CLOSE(FOUT)
END.
```

irandom

ROUTINES

```
func rrandom(min,max:real):real;  
func irandom(min,max:integer):integer;
```

keys

IMPORTANT KEYBOARD KEYS ON THE R65 SYSTEM

CTRL-l	split/full screen graphics
CTRL-r	print all on
CTRL-t	print all off

CTRL-<return>	clear to end of screen
CTRL-<right>	insert char
CTRL-<left>	delete char
CTRL-<down>	scroll down
CTRL-<up>	scroll up

SHIFT-CTRL	is required on UTM/Debian and on Macintosh
------------	---

ladders

LADDERS – Climb ladders and avoid a bouncing ball

You are entering a building with several floors at the bottom left corner. There are ladders to climb up and down between floors. There are also holes, over which you can jump. The exit is at top right corner.

You win the game if you reach the exit, but there is also a bouncing ball coming from the exit. The ball can fall through and jump over the holes. You loose if the ball hits you.

Command line parameter:
/D endless demo mode

Keys

left/right	move to the left/right
up/down	climb ladders up/down
ESC	quit the game

You jump automatically over holes if you are fast enough.

ledlib

LIBRARY LEDLIB;

MEMORY

LEDREG=\$1432: array[7] of char&;

ROUTINES

proc _ledstring(s:cpnt);

proc _ledstop;

proc _ledhex(d,p,digits: integer);

proc _ledbyte(d: integer);

mathlib

LIBRARY MATHLIB;

CONSTANTS

PI = 3.14159;
E = 2.71828;

ROUTINES

```
func fabs(x:real):real;  
func sqrt(n:real):real;  
func cos(x:real):real;  
func sin(x:real):real;  
func tan(x:real):real;  
proc writeflo(r:real; fl:integer);  
proc writef0(r: real; fl, d:integer;  
  centered:boolean);  
proc __wrfix(r:real; fl, d:integer);  
func readflo(f:file):real;  
func ln(r:real):real;  
func exp(x:real):real;  
func log(x:real):real;
```

nroff

NR0FF(1)

NAME

=====

NR0FF – text formatter for R65

SYNOPSIS

=====

NR0FF filnam[:sub[.drv]] [linewidth]

Default line width is 40.

DESCRIPTION

=====

NR0FF formats plain text files using simple dot commands. Lines starting with a dot are formatting requests. All other lines are normal text and are formatted using fill mode.

TEXT MODES

=====

Fill mode Words are wrapped to fit the current line width.

No fill mode Text is printed exactly as entered.

.fi Enable fill mode.

.nf Enable no fill mode.

HEADINGS

=====

.TH Manual header line.

.SH Section heading. Uses underline or inverse video depending on width.

.SS Subsection heading.

PARAGRAPHS

=====

.PP Start new paragraph.

.br Line break.

.sp [n] Insert blank lines.

INDENTATION

=====

.RS Increase indentation.
 .RE Restore indentation.
 .IP labelIndented paragraph. Label followed by
 text with hanging indent.

TABS

=====

.ta n1 n2 ...Set tab stops.
 Tabs are expanded during output.
 If no tabs are defined, default spacing is 8
 cols.

TEXT STYLE

=====

.B text Bold line. On 40 columns inverse video
 is used.

INPUT RULES

=====

Maximum input line length depends on buffer
 size.
 Control characters except TAB are ignored.

OUTPUT WIDTH

=====

Line width may be given as optional parameter.
 Example NROFF MANUAL:B,1 80
 Widths above 40 use printer style headings.

TABLES

=====

Tables are created using no fill mode and tabs.
 Example:

COL1	COL2	COL3
====	====	====
AA	BBB	CCCC

LIMITATIONS

=====

No macros or registers.
 No escape sequences.
 Formatting is single pass.

AUTHOR

=====

Written for the R65 system.
Implementation by R. Richarz and ChatGPT.

END

===

End of NR0FF manual. □

□

paint

PAINT image – paint in graphics canvas

Paint with the following keys:

right arrow	_move cursor right
left arrow	_move cursor left
up arrow	_move cursor up
down arrow	_move cursor down
C	clear image
p	paint a dot at cursor position
L	drawing line mode
R	drawing rectangle mode
S	drawing character mode
return	_draw object and exit mode
esc	exit mode without drawing
W	write image to disk
Q	write image to disk and quit
K	kill program without writing to disk

pcodes

PCODES filename[disk,from,to]

Display P-codes of a Pascal Program.
Pascal source code and object code must be
on the same drive.

If the program has been compiled with the /L
option (line numbers), pcode displays the
source line together with the pcodes for this
line, and pcodes of a specified range of
lines (from/to) can be displayed. Use SETB
and RESETB to set breakpoints.

pedit

PEDIT filename

ESC	Enter command
t	Go to top of file
b	Go to bottom of file
l n	Go to line n
f	Find
a	Find again
z	Clear marks
c n	Mark lines of copy
p	Paste copied lines
m	Move copied lines
d n	Delete n lines
w	Write file
q	Write file and quit
k	Kill (quit without writing)
?,h	Help

plotlib

LIBRARY PLOTLIB;

CONSTANTS

```
XSIZE=223;  
YSIZE=117;  
XWORDS=28;  
WHITE=0;  
INVERSE=1;  
BLACK=2;  
PLOTDEV=@128;
```

MEMORY

```
KEYPRESSED=$1785: char&;
```

VARIABLES

```
_xcursor, _ycursor: integer;
```

ROUTINES

```
proc _delay10msec(time:integer);  
func _syncscreen;  
proc _grinit;  
proc _grend;  
proc _cleargr;  
proc _fullview;  
proc _splitview;  
proc _plot(x,y,c:integer);  
proc _move(x,y:integer);  
proc _draw(x,y,c:integer);  
proc _plotmap(x,y,m:integer);  
proc _waitforKEY;  
proc _plotif(x,y,mode: integer);  
proc _circle(cx,cy,r,mode: integer);  
proc _rectangle(x, y, sx, sy, mode: integer);
```

R65

R65 QUICK REFERENCE

=====

MONITOR COMMANDS

=====

DIR Directory listing.
COPY Copy file between drives.
DELETE Delete a file.
EDIT Edit text file.
LOAD Load file to memory.
STORE Store memory sektion to disk.
GO Execute machine code in memory.
DIS Disassemble memory.
REG Show CPU registers.
DATE Show or set date.
FLOPPY Select disk image.
IMPORT Import host file.
EXPORT Export file to host.
RUN Run a maschine code program.

PASCAL

=====

COMPILE Compile Pascal program.
EDIT Edit a text file.
name Execute Pascal program name.
LIB Load Pascal library.

NR0FF

=====

NR0FF Format text document.
.TH Manual header.
.SH Section heading.
.SS Subsection heading.
.PP New paragraph.
.IP Indented paragraph.
.nf / .fi No fill / fill mode.
.ta Set tab stops.

SCREEN CONTROL

=====

CT-E Clear line.
CT-RTN Clear to end of display.

CT-RGT	Insert character.
CT-LFT	Delete character.
CTRL-A	Cursor home.
CT-UP	Roll up.
CT-DN	Roll down.

ralib

LIBRARY RALIB;

CONSTANTS

```
FREAD=$00;  
FWRITE=$20;  
FNEW=$30;
```

ROUTINES

```
func _uppercase(ch:char):char;  
func _attach(fname:array[15] of char;  
    cyclus,drive,operation,size,start:integer;  
    subtype:char):file;  
func _getsize:integer;  
proc _getword(device:file; address:integer;  
    var word:integer);  
proc _putword(device:file; address:integer;  
    word:integer);  
proc _getreal(device:file; address:integer;  
    var rvalue:array[1] of %integer);  
proc _putreal(device:file; address:integer;  
    rvalue:array[1] of %integer);
```

restore

RESTORE filename

Restore deleted files(s).
Wildcars allowed.

Use DIR /D to display deleted files

snap

SNAP image – save a snapshot in image

Additional commands for images:

SHOW image – show image

PAINT image – paint image

snippets

```
{ check for options }
if ARGTYPE[_carg]='s' then begin
  sortit  := option('S');
  full    := option('F');
  if not (sortit or full) then begin
    writeln('option must be /S or /F or /FS');
    exit;
  end;
end;

{ find file entry with best match }
writeln;
if not _bestmatch then begin
  writeln(CUP,INVVID,'File not found',NORVID);
  exit;
end;
writeln;
```

spritelib

LIBRARY SPRITELIB;

CONSTANTS

```
NSPRITES      = 64;
FRAMES        = 3;
S_CLEAR       = 0;
S_STANDING    = 1;
S_RIGHT       = 9;
S_LEFT        = 32;
S_UP          = 14;
S_DOWN        = 35;
S_JUMPR       = 38;
S_JUMPL       = 41;
S_TURN        = 19;
S_CIRCLE      = 45;
S_BALL        = 46;
S_SMILEY      = 47;
S_PIECE_WHITE = 51;
S_PIECE_BLACK = 52;
S_GEMS        = 53;
```

VARIABLES

```
sprites: array[256] of integer;
```

ROUTINES

```
proc _readsprites(drv: integer);
proc _showsprite(x, y, index: integer);
```

starship

STARSHIP – A Starship Game

You command a starship while enemy ships attempt to destroy you. Your ship has several systems: scanner, shield, phaser, and warp engine. Each system consumes energy when used and can also be damaged.

If a system is damaged, its maximum available energy is reduced by the amount of damage. Energy is restored slowly over time, but one selected system can be recharged more quickly. Damaged systems can also be repaired.

The scanner detects enemy ships within its range, which depends on the scanner's energy level and repair status. Enemy positions are displayed on a radar-like screen, showing both angle and distance.

The phaser has a much shorter range, which also depends on the available energy and the repair status of the phaser. It is fired using the angle obtained from the scanner.

The shield becomes less effective when damaged and must be lowered before firing the phaser.

The condition of the warp engine determines the maximum distance the ship can warp. Both angle and distance must be specified.

The game ends when all enemy ships have been destroyed or when your own starship is destroyed.

striolib

LIBRARY STRIOLIB;

CONSTANTS

 NAMESIZE = 15;

ROUTINES

```
proc _checkfilerr;  
proc _sgetstring(string: cpnt; var scarg: integer;  
                  var default: boolean);  
proc _ssetsubtype(sname: cpnt; subtype:char;  
                  force: boolean);  
proc _strfio(name:cpnt; cyclus, drive: integer);  
proc _getdiskname(s: cpnt; drive: integer);  
proc _change_disk(s: cpnt; drv: integer);  
proc _s_runprog(name: cpnt; cyc, drv: integer);  
proc _s_chainprog(name: cpnt; cyc, drv: integer);
```

strlib

LIBRARY STRLIB;

CONSTANTS

```
    STRSIZE=64;  
    ENDMARK=chr(0);
```

ROUTINES

```
proc _runerr(e:integer);  
func _new: cpnt;  
proc _release(s: cpnt);  
func _strlen(strin:cpnt):integer;  
proc _strcpy(strin, strout:cpnt);  
proc _stradd(strin,strinout:cpnt);  
func _strcmp(s1,s2:cpnt):integer;  
func _strpos(ch:char; s1:cpnt;  
            start:integer): integer;  
func _strread(f:file; s:cpnt;  
            var ateof0:boolean):integer;  
proc _hexstr(d:integer; s:cpnt);  
proc _strdelc(pos:integer; s:cpnt);  
proc _strinsc(ch:char;pos:integer;s:cpnt);  
proc _intstr(n:integer;s:cpnt;fsize:integer);
```


syslib

LIBRARY SYSLIB;

CONSTANTS

```
TAB8      = chr(9);
HOM       = chr(1);
CSC       = chr($11);
LF        = chr($a);
FF        = chr($c);
CR        = chr($d);
EOF       = chr($7f);
CUP       = chr($1a);
CLRLIN    = chr($17);
NORVID    = chr($0b);
INVVID    = chr($0e);
PRTON     = chr($12);
PRTOFF    = chr($14);
TOPMEM    = $c780;
MAXINT    = $7fff;
INPUT     = @0;
OUTPUT    = @0;
KEY       = @1;
PRINTER   = @1;
ECEXPRT   = 1;
ECIMPORT  = 2;
ECEDIT    = 3;
EC_CLR_PRTOUT = 9;
```

MEMORY

```
RUNERR    = $000c: integer&;
FILERR    = $00db: integer&;
ENDSTK    = $000e: integer;
IOCHECK   = $0023: boolean&;
CURPOS    = $00ee: integer&;
```

VARIABLES

```
_day:     integer;
_month:   integer;
_year:    integer;
```

ROUTINES

```
func _emulator(i:integer): integer;
func _getbcd(address: integer): integer;
proc _getdate;
```

```
proc _prtdat(device: file);  
func _abs(x: integer): integer;  
func _mod(x,n: integer): integer;  
proc _tab(pos: integer);  
proc _abort;  
proc _prtext8(device: file; text: array[7] of char);  
proc _prtext16(device: file; text: array[15] of char);  
func _random: integer;  
proc __wrintf(value, width: integer);  
func _readint(f:file; var n:integer): boolean;  
func _isesc:boolean;
```

table

graph

display the graph of function values
stored in table

table

make a table for display with graph

teklib

LIBRARY TEKLIB;

CONSTANTS

```
MAXX = 1023;
MAXY = 780;
MAXCOLUMNS = 74;
MAXLINES    = 35;
SOLID       = 1;
DOTTED      = 2;
DOTDASH     = 3;
SHORTDASH   = 4;
LONGDASH    = 5;
PLOTTER     = @1;
T_HALF      = 0;
T_FULLLV    = 1;
T_GAMING    = 2;
```

VARIABLES

```
_xs,_ys: integer;
```

ROUTINES

```
proc _delay10msec(time:integer);
proc _clearscreen;
proc _starttek(mode: integer);
proc _endtek;
proc _startdraw(x1,y1:integer);
proc _draw(x2,y2:integer);
proc _enddraw;
proc _drawvector(x1,y1,x2,y2:integer);
proc _drawrectangle(x1,y1,x2,y2:integer);
proc _moveto(x1,y1: integer);
proc _setchsize(size:integer);
proc _setlinemode(type:integer);
```

timelib

LIBRARY TIMELIB;

VARIABLES

tenmillis,seconds,minutes,hours,
diffftenmillis: integer;

ROUTINES

proc gettime;
proc prttime(device: file);
func timediff: real;

view

VIEW filename

ESC, q	Quit
CUP, CDOWN	Scroll up/down
CLEFT, CRIGHT	Scroll 12 lines up/down
/, f	find string
n	next hit
?, h	help

wildlib

LIBRARY WILDLIB;

CONSTANTS

```
    NAMESIZE    = 15;  
    NUMENTRIES  = 255;
```

ROUTINES

```
proc _findentry(var nm:array[NAMESIZE] of char;  
    drv:integer;var ent: integer;  
    var fnd,lst:boolean);  
proc _sfindentry(name: cpnt; drv:integer;  
    var ent: integer;  
    var fnd,lst: boolean);  
proc _writename(nm1:array[15] of char);
```